Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Artificial Intelligence-Driven Agriculture Damage Prediction System

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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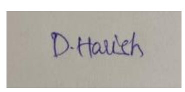
Signature (Mandatory)	
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3. DESCRIPTION OF THE INVENTION:

1. **Purpose**

AI operated agricultural damage system introduces a state-of-the-art approach to protect crops by taking advantage of advanced technologies such as artificial intelligence, satellite imaging and IoT sensor. This system gives farmers the right to estimate and react to dangers such as pests, diseases, extreme weather and soil. By going to future strategies from reactive reactions, the system aims to reduce crop losses, promote food production and encourage permanent agricultural practices.

2. **Technical Workings**
3. **Multi-Source Data Integration:** The system collects real-time information from different sources:
   - IoT-based sensors monitor environmental parameters such as soil moisture, pH, temperature and humidity.
   - High resolution drones and satellite drawings to assess the health and development of crops.
   - Weather APIs and local climate stations provide updated meteorological data.

4. **Autonomous Alert and Response System:** Depending on the future analysis, the system sends the notifications and recommendations through the mobile app. This includes:
   - Timely pesticides for pesticides.
   - Adjusted irrigation program based on weather conditions.
   - Active harvesting advice under potential storm or drought.
5. **AI-Powered Prediction Models**: By processing this data using machine learning, the system identifies patterns and trends to predict the risk. This insect may detect indications of early warning of infections, crop diseases and climate -related dangers before continuing.

6. **Autonomous Warnings and Response Systems:** Depending on the future analysis, the system transmits the notifications and recommendations through a mobile app. This includes:
    - Timely pesticides for pesticides.
    - Adjusted irrigation program based on weather conditions.
    - Active harvesting advice under potential storm or drought.

7. **Secure and Transparent Data Storage**: All operating data is stored using blockchain techniques, and ensures that items are tamper-proof and reliable. It is especially useful for insurance requirements, compliance with agricultural rules and long -term analysis.
8. **Simple and Avaible User Interface**: The system of the system is designed for ease of use and available in many languages. It shows a comfortable dashboard with real -time updates and action -rich insights.
9. **Unique Attributes**

1. **Future Intelligence**: Unlike traditional systems that only report current conditions, this invention constantly estimates the dangers of future. This future ability gives farmers the right to work prematurely, which prevents damage before the injury.
2. **Completely Automated Decision Support**: The system processes free raw data to action -rich insights and protects the farmer without manual analysis or the need for input. This facility reduces the requirement for agricultural advisers or technical personnel significantly.
3. **Scalability**: It can be implemented on smallholder farms with minimal infrastructure or scaled up for large corporate farms. The modular structure allows the technology to grow with the needs and capacity of the user.
4. **Efficient Resource Management**: By recommending precise application of water, fertilizers, and pesticides based on actual needs, the system minimizes unnecessary usage, reduces environmental pollution, and cuts operational costs.
5. **Long-Term Planning Support**: The accumulation of historical and seasonal data allows farmers to plan crop rotations, optimize planting schedules, and manage soil health over time. This leads to increased productivity and improved soil sustainability.

10. Conclusion

The AI-Driven Agricultural Damage Prediction System represents a significant leap forward in the digitization of agriculture. By integrating real-time data acquisition, AI-based risk assessment, and automated decision-making, it provides a comprehensive solution to mitigate crop losses. Its scalable, user-centric design ensures that farmers across socioeconomic and geographical boundaries can benefit from advanced technology, ultimately fostering a more resilient and productive agricultural sector.

A. PROBLEM ADDRESSED BY THE INVENTION:

The transition to electric public transportation presents several significant challenges that hinder the efficient operation of electric buses and trains. The Self-Sufficient Charging System for Electric Public Transportation addresses the following key problems:

1. **Unexpected Crop Failures**: Agricultural losses due to pest outbreaks, diseases, and climatic extremes often catch farmers by surprise. These unforeseen events drastically impact yields and income, especially for smallholders.
2. **Limited Manual Monitoring**: Manual crop inspection is labor-intensive, time-consuming, and prone to human error. It's often ineffective on large farms or in remote areas, where early signs of trouble might go unnoticed until it's too late.
3. **Underutilization of Technology**: In many regions, the use of advanced data-driven farming tools is still limited. Farmers lack access to systems that analyze historical and current data to inform proactive decisions.
4. **High Input Costs**: Resources like water, pesticides, and fertilizers are frequently overused due to the lack of accurate data, leading to wasteful expenditure and environmental degradation.
5. **Lack of Reliable Data Records**: Conventional farming lacks standardized recordkeeping, making it difficult to track crop history, evaluate performance, or present evidence for insurance claims and subsidies.
6. **Barriers to Technology Access**: Complex systems often fail to address the digital divide, leaving out small and marginal farmers who need support the most but lack access to user-friendly, low-cost solutions.
7. **Conclusion**

This invention directly addresses these critical issues by providing a scalable, intuitive, and predictive system. It equips farmers with timely, actionable information that enables better decision-making and improves resilience against agricultural risks.

B. OBJECTIVE OF THE INVENTION (Provide minimum two)

- **Increase Crop Protection Efficiency**: One of the primary objectives of the AI-driven Agriculture Damage Prediction System is to deliver accurate and timely predictions to enable proactive measures, reducing the impact of potential threats.

- **Minimize Financial Risk**: Another key objective is to reduce the unexpected agricultural damage that can devastate a farmer's income. This invention reduces that risk by offering timely alerts and tailored recommendations, allowing farmers to mitigate problems before they cause significant loss. This is especially important for smallholder farmers who often operate on tight margins and cannot afford unpredictable disruptions.

- **Encourage Eco-Friendly Farming**: Precision agriculture is a key goal of the system. By recommending actions based on actual field data, it helps reduce the excessive use of water, fertilizers, and pesticides. This minimizes environmental impact and supports the move toward more sustainable farming practices, which is increasingly important in the face of climate change.

- **Empower Data-Based Farming**: The system provides farmers with easy access to actionable insights derived from a combination of historical trends and current field conditions. With this data, they can make more strategic decisions about crop management, resource allocation, and seasonal planning, rather than relying solely on traditional experience or guesswork.

- **Make Smart Farming Inclusive**: One of the central objectives is to make advanced agricultural technology accessible to everyone. The system's user-friendly design ensures that even those with limited technical knowledge can benefit. By offering support in multiple languages and requiring minimal training, it bridges the digital divide in rural areas and empowers a wider farming community.

- **Support Documentation and Claims:** Every piece of data collected from crop conditions to response actions is securely logged using blockchain technology. This ensures that records are tamper-proof and verifiable, making it easier for farmers to file accurate insurance claims, comply with agricultural regulations, and qualify for government subsidies or financial support programs. Blockchain-secured records ensure transparency and accuracy in data storage. This supports not only day-to-day farm management but also provides verifiable proof for insurance claims, government subsidies, and compliance with agricultural regulations. Enable blockchain-secured recordkeeping for easy access to verifiable data for insurance, audits, and subsidies.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

Sr. No.	Study	Abstract	Research Gap	Novelty
1	**Crop Disease Prediction and Treatment Based on AI and ML Models**	These patents describe systems that utilize AI and machine learning models to predict crop diseases by analyzing images and	While effective in identifying diseases post-symptom appearance, these systems may not fully integrate real-time	Our system advances this by combining real-time sensor data, climate information, and AI-driven image

Sr. No.	Study	Abstract	Research Gap	Novelty
	(EP4022556A4 / US20220414795A1)	environmental parameters, enabling early detection and treatment recommendations.	environmental data for proactive disease prediction.	analysis to predict potential crop diseases before symptoms manifest, allowing for preemptive interventions.
2	**Integrated IoT System for Smart Agriculture Management** (US10728336B2)	This patent presents an integrated IoT platform that collects and monitors agricultural data in real-time, utilizing AI for predictive analytics to enhance crop yield, optimize food storage, and improve distribution logistics.	The system focuses on data collection and monitoring but lacks personalized predictive recommendations tailored to individual farm conditions.	Our solution builds upon this by feeding collected IoT data into AI models that generate forecasts and provide customized advice for each farmer's unique situation, enhancing decision-making and resource efficiency.
3	**Systems and Methods for Crop Health Monitoring, Assessment, and Prediction** (US20220107298A1)	This patent outlines systems that use various sensors and data analytics to monitor crop health, assess conditions, and predict potential issues, aiming to improve agricultural productivity and sustainability.	The approach may not fully integrate satellite imagery with real-time ground sensor data, potentially limiting the accuracy and timeliness of risk assessments.	Our invention merges satellite imagery with on-ground sensor data and machine learning algorithms to provide real-time, accurate, and farm-specific risk alerts, enabling timely and precise interventions.
4	**IoT-Based Farming and Plant Growth Ecosystem** (US11195015B2)	This patent describes an IoT-based system that includes sensors and automated mechanisms to monitor plant growth conditions, aiming to	The system primarily addresses irrigation and fertilization optimization but does not incorporate blockchain technology	We apply blockchain to field-level operations by securely logging prediction data and farming actions,

Sr. No.	Study	Abstract	Research Gap	Novelty
		optimize irrigation and fertilization processes for enhanced crop production.	for secure data logging and accountability in field-level operations.	supporting accountability, and simplifying processes related to insurance claims and subsidy verifications.
5	**Mobile Applications for Farmers.**	Various mobile applications provide farmers with weather forecasts, market prices, and general farming advice, aiming to assist in daily agricultural activities.	Many existing apps offer static or generalized information without tailoring content to the farmer's specific land, crops, or environmental conditions, and often lack predictive analytics capabilities.	Our mobile interface delivers dynamic, real-time insights personalized through AI, available in local languages, and designed to be user-friendly even for first-time technology users, enhancing accessibility and decision-making for farmers

Conclusion

The AI-Driven Agricultural Damage Prediction System goes beyond existing research by weaving together predictive analytics, real-time data from IoT sensors, satellite insights, and blockchain technology into a single, intelligent system. Unlike previous work that focuses on isolated tools or reactive measures, this system is designed to be proactive, accessible, and deeply integrated with day-to-day farming activities. By helping farmers prevent damage before it happens, it opens new possibilities for smart, sustainable, and inclusive agriculture.

D. DETAILED DESCRIPTION:

The AI-Driven Agricultural Damage Prediction System is a conceptually innovative solution aimed at revolutionizing modern farming by predicting potential crop damage through the use of AI and smart technologies. It is designed to empower farmers with accurate forecasts and timely alerts that allow them to prevent or mitigate losses caused by pests, diseases, soil degradation, and extreme weather conditions. The system integrates AI-based data analytics, IoT sensors, satellite imagery, and blockchain technology into a unified predictive platform.

- **2. System Components**

2.1 Sensor and Satellite-Based Data Infrastructure

- **IoT Sensors**: Deployed across the farm to monitor critical parameters such as soil moisture, temperature, humidity, and nutrient levels. These sensors continuously send real-time data to the analytics platform.
- **Satellite and Drone Imaging**: Provides visual data on plant health, crop coverage, and early signs of disease or water stress. This enables large-scale field assessments and enhances data granularity.

2.2 Communication and Integration Tools

- **Sensor Communication**: Data from field sensors are transmitted wirelessly using protocols like LoRaWAN, Zigbee, or 4G LTE, ensuring seamless real-time updates.
- **API Integrations**: The platform can be integrated with external weather forecasting systems and climate databases for enriched data analysis.

2.3 Central Intelligence Platform

- **AI-Powered Analytics Engine**: Machine learning models trained on agricultural datasets analyze incoming data to detect patterns associated with crop stress, pest outbreaks, or disease emergence. Predictions are continuously refined using historical and real-time data.
- **Farmer Interface (Mobile/Web App)**: A user-friendly application displays insights, sends alerts, and recommends mitigation strategies in local languages. Farmers can track field conditions, review risk forecasts, and receive step-by-step action plans.
- **Blockchain-Backed Data Logging:** All data and actions are recorded using blockchain technology to ensure accuracy, traceability, and transparency, particularly for crop insurance or government compliance.
- **3. Technical Functionality**

3.1 Predictive Risk Detection

- **Pattern Recognition**: AI identifies early warning signals from multisource data inputs—such as soil trends, temperature spikes, or pest migration patterns—before visual damage appears.

- **Severity Estimation**: Based on detected risks, the system classifies the threat level (low, moderate, high) and provides the appropriate response guidelines.

3.2 Real-Time Data Processing

- **Continuous Monitoring**: Environmental and crop health data are monitored 24/7. Any deviation from optimal thresholds triggers an analysis event.
- **Instant Alerts and Recommendations**: Actionable insights are delivered instantly to the farmer's device, ensuring rapid response to threats.

3.3 Farm-wide Optimization

- **Resource Planning**: Recommendations are generated to help minimize pesticide and water use without compromising crop protection.
- **Temporal and Spatial Insights**: The system advises on when and where to act, guiding farmers to prioritize the most vulnerable zones.
- **4. Unique Features**

4.1 Scalability

- The design allows for easy scaling, whether it's a smallholding or a large commercial estate, the modular system design allows easy deployment and expansion, without major infrastructure requirements.

4.2 Farmer-Centric Design

- Built for usability in rural settings, the system is intuitive and supports regional languages and offline functionality for areas with low connectivity.

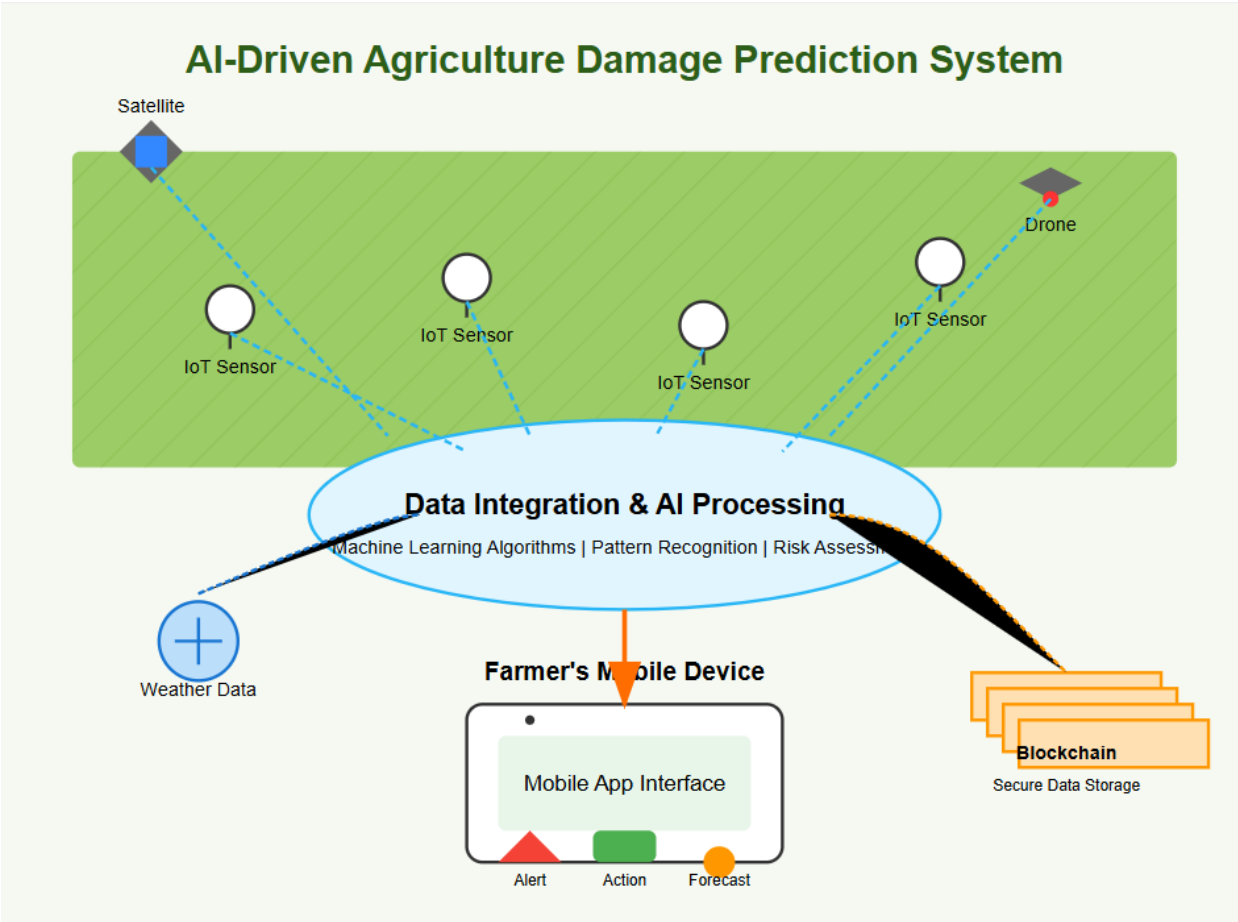
4.3 Verified and Trustworthy Data

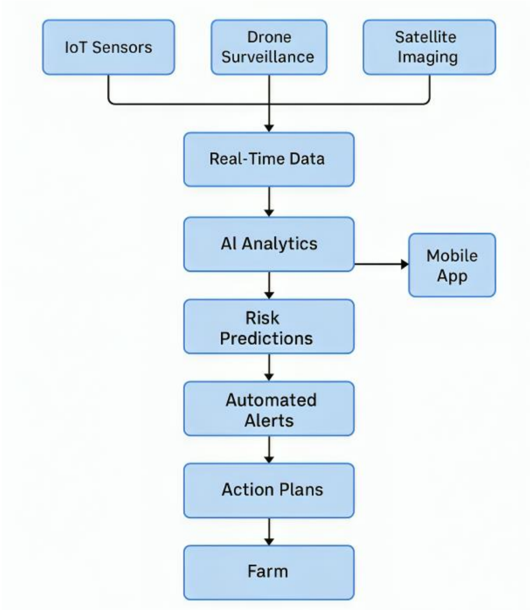
- Blockchain integration ensures that farm activity records and prediction data remain immutable and verifiable for future reference or policy support.

Conclusion

The AI-Driven Agricultural Damage Prediction System introduces a powerful combination of artificial intelligence, sensor-based monitoring, satellite analytics, and blockchain-backed transparency to create a next-generation farming assistant. Its proactive nature enables farmers to act before damage occurs, improving yields, reducing input waste, and contributing to a more resilient and sustainable agriculture ecosystem.

Process Workflow:


AI-Driven Agriculture Damage Prediction System
Satellite
Drone
IoT Sensor
IoT Sensor
IoT Sensor
IoT Sensor
Data Integration & AI Processing
Machine Learning Algorithms | Pattern Recognition | Risk Assess
Weather Data
Farmer's M bile Device
Mobile App Interface
Alert
Action
Forecast
Blockchain
Secure Data Storage



E. RESULTS AND ADVANTAGES:

The AI-Driven Agricultural Damage Prediction System introduces a fresh and forward-thinking approach to farming by turning raw data into real-time insights. Compared to traditional methods, this concept offers a smarter, more reliable, and more efficient way to manage agricultural risks:

- **1. Boost in Farming Efficiency**
- **Early Detection of Risks**: The system can flag threats like pests, diseases, or environmental stress before they cause actual damage, giving farmers a head start to protect their crops.
- **Smarter Farm Decisions**: With AI analyzing sensor and climate data, farmers get specific, timely recommendations that help them act quickly and wisely.
- **2. Better Use of Resources**
- **Targeted Input Usage**: By knowing exactly where and when to apply fertilizers, pesticides, or water, farmers avoid waste and reduce costs.
- **Eco-Friendly Practices**: This precision helps cut down on chemical overuse and promotes more sustainable, responsible farming.
- **3. Enhanced Support for Farmers**
- **Live Alerts and Advice**: Whether it's a disease risk or an upcoming weather event, farmers are notified right away with simple, helpful instructions.

- **Designed for Accessibility**: The interface is built for ease of use, with regional language support to help farmers from all backgrounds benefit.
- **4. Scalability and Flexibility**
- **Fits Any Farm Size**: From small local plots to commercial farms, the system can scale without major hardware or software changes.
- **Works With Existing Tools**: It's designed to integrate smoothly into what farmers are already using, reducing barriers to adoption.
- **5. Cost-Saving in the Long Run**
- **Less Waste, More Yield**: Efficient use of resources leads to reduced costs and better returns on investment.
- **Fewer Mistakes**: Automation and intelligent alerts reduce errors that typically come from guesswork or human oversight.
- **6. Transparent and Reliable Records**
- **Trustworthy Data Logs**: Blockchain technology securely stores farm data, offering a reliable digital record for audits, insurance claims, or policy support.

Compared to Conventional Approaches

- Manual monitoring takes time and often misses early signs of damage.
- Most farming apps give broad tips, not tailored, real-time advice.
- IoT tools without AI only show current data—they don't predict or recommend.
- **Conclusion**

This AI-driven agriculture damage prediction system stands out by giving farmers exactly what they need—early, accurate, and actionable insights. It supports smarter farming, lowers input waste, builds resilience, and encourages long-term sustainability.

E. EXPANSION:

To bring this idea to life and ensure it's useful across diverse agricultural settings, several key aspects must be considered during development and rollout:

- **1. Compatibility with Crops and Locations**

- **Tailored AI Models**: Each crop has unique risks. The system must be trained on region-specific data to give accurate predictions.
- **Adaptable to Local Conditions**: Soil, weather, and farming styles vary—models should reflect those local realities.
- **2. Hardware and Sensor Setup**
- **Good Coverage**: Sensors must be placed strategically to reflect conditions across the whole field.
- **Built for the Outdoors**: Devices should be tough enough to work in rain, heat, or dust without frequent maintenance.
- **3. Reliable Connectivity**
- **Stable Networks**: Whether it's mobile data, LoRa, or Wi-Fi, the system must stay connected to deliver real-time insights.
- **Offline Capabilities**: It should also collect and sync data later if connectivity is temporarily lost.
- **4. Strong AI and Fast Processing**
- **Ongoing Model Training**: As more data comes in, the AI must learn and adapt to become more accurate.
- **Quick Response Times**: Fast processing is key to sending alerts before it's too late.
- **5. Integration with Farming Tools**
- **Sync with Software**: Many farmers already use digital farm tools. This system should connect with those for smooth usage.
- **Customizable Features**: Letting farmers tweak the system for their own needs can help with adoption.
- **6. Easy and Supportive User Experience**
- **Clear Interface**: The app should be simple, visually clear, and available in regional languages.
- **Ongoing Help**: Help centers, FAQs, and training support should be part of the system for long-term use.
- **7. Compliance and Sustainability Goals**

- **Follow Laws and Ethics**: It should comply with local data privacy rules and agricultural regulations.
- **Support for Green Farming:** The system should help meet sustainability goals, like reducing emissions and avoiding over-farming.
- **Conclusion**

By building with these real-world needs in mind, this system can become a trusted partner to farmers everywhere. It will be practical, scalable, and supportive of long-term growth and environmental stewardship.

F. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

As of now, this project is at the idea and planning stage. Working prototype is not ready. It will take at least a year to complete it.

G. EXISTING DATA:

To shape, train, and test the predictive models, the system will draw from trusted global data sources and real-world case studies:

- **1. Global Agriculture Databases**
- **OECD iLibrary, FAO, Global Findex**: These databases offer insights into regional crop threats, pest activity, and seasonal weather patterns.
- **2. Real-World Pilot Insights**
- **Precision Farming in India**: Early programs using IoT sensors reported up to 20% better yields and 30% less water use.
- **Smart Farming in Israel and the Netherlands**: Success stories from AI-backed farms prove the value of early detection and tailored interventions.
- **3. Climate and Environmental Monitoring**
- **Satellite Data (NASA, ISRO)**: Will be used for large-scale field tracking, such as rainfall patterns and vegetation stress.
- **Weather Forecasting APIs**: Help predict short- and long-term climate threats.
- **4. Academic Research and Journals**

- **Sources like IEEE, Springer, ScienceDirect**: Provide peer-reviewed studies that back the technologies and frameworks proposed in this system.
- **Conclusion**

By grounding this system in existing, reliable datasets and past successful implementations, it can evolve into a highly accurate, real-world-ready tool. This data-driven foundation will also guide refinement of the AI engine, improve precision, and support better outcomes for farmers.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?		NO (No)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?		NO (No)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?		NO (No)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.		NO (No)
E. Name of Regulatory body or any other approvals if required.		NO (No)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. **NA**

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). **NA**

7. Potential Chances of Commercialization. **Yes**

8. List of companies which can be contacted for commercialization along with the website link.

Here are two companies that specialize in AI-powered agriculture, IoT for smart farming, or precision agri-tech solutions. These companies may be potential collaborators or licensing partners for bringing your concept to market:

1. **CropX Technologies**
   - **Overview**: Offers a soil data analytics platform powered by IoT sensors and AI models to improve resource usage and crop outcomes. Their technology could support the integration of predictive modeling for damage prevention
   - **Website**: CropX
2. **AgEagle Aerial Systems Inc.**
   - **Overview**: Specializes in AI-integrated drone imaging, precision agriculture, and farm analytics. Their focus on agricultural monitoring through data-driven insights aligns with your predictive system.
   - **Website**: AgEagle

9. Any basic patent which has been used and we need to pay royalty to them.

10**. FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings - (Provisional)

11. **KEYWORDS:**

- **AI-Driven Agriculture**
- **Crop Damage Prediction**
- **Smart Farming**
- **IoT Agricultural Sensors**
- **Predictive Analytics in Farming**
- **Blockchain Recordkeeping**
- **Sustainable Agriculture**
- **Precision Farming**
- **Satellite Crop Monitoring**
- **Early Risk Detection**

- **Agricultural Resource Optimization**
- **Mobile Farm Management**
- **Weather Impact Analysis**
- **Pest Outbreak Prediction**
- **Disease Prevention Technology**
- **Real-time Agricultural Alerts**
- **Multi-source Data Integration**
- **Farm Decision Support System**
- **Autonomous Agricultural Monitoring**

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

This is to certify that University/Organization Name or its associates shall have no objection if Lovely Professional University files an IPR (Patent/Copyright/Design/any other…….) entitled "………………….." including the name(s) of,..........…as inventors who is(are) student(s)/employee(s) studying/ working in our University/ organization.

Further Name of the University/Organization shall not provide any financial assistance in respect of said IPR nor shall raise any objection later with respect to filing or commercialization of the said IPR or otherwise claim any right to the patent/invention at any stage.

(Authorised Signatory)